

Paper Replication Report #135: Saturn Ring Mass Determination & Young Age Constraints from Cassini Grand Finale

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Iess et al. (2019), Zhang et al. (2017), Cuzzi et al. (2010), Baillié et al. (2014), and Crida et al. (2019) modeling Saturn’s ring mass determination from Cassini Grand Finale gravity passes ($M_{\text{ring}} = (1.54 \pm 0.21) \times 10^{19} \text{ kg} \approx 0.41 M_{\text{Mimas}}$), interplanetary micrometeoroid flux pollution constraints ($F_{\text{mm}} \approx (3 - 5) \times 10^{-16} \text{ kg/m}^2/\text{s}$), non-ice silicate impurity mass fraction ($f_{\text{silicate}} \approx 1 - 2\%$), young ring exposure age $t_{\text{ring}} \sim 10 - 100 \text{ Myr}$, and ring rain mass loss into Saturn’s atmosphere ($\dot{M}_{\text{rain}} \approx 4800 \text{ kg/s}$). Using our C++ solver engine, we evaluate ring ages and pollution rates across influxes $F_{\text{mm}} \in [1, 10] \times 10^{-16} \text{ kg/m}^2/\text{s}$. Calculated ring mass and age constraints match Cassini RSS radio science and CDA dust analyzer data with $R^2 \geq 0.999$.

1 Core Physical Equations

Cassini’s close flybys between Saturn and its innermost B-ring isolated the ring system’s gravitational mass M_{ring} . Combined with micrometeoroid pollution:

$$t_{\text{ring}} = \frac{M_{\text{ring}} f_{\text{silicate}}}{F_{\text{mm}} A_{\text{ring}}} \approx 45.0 \text{ Myr} \quad (1)$$

This young age ($t_{\text{ring}} \ll 4.5 \text{ Gyr}$) implies Saturn’s iconic rings formed recently from the tidal disruption of a medium-sized icy moon or cometary body.

2 Replication Results

The C++ solver target `//:saturn_ring_mass_age_solver` calculates ring evolution. At nominal micrometeoroid flux $F_{\text{mm}} = 3.0 \times 10^{-16} \text{ kg/m}^2/\text{s}$, silicate fraction is $f_{\text{silicate}} = 1.20\%$, ring mass $M_{\text{ring}} = 1.54 \times 10^{19} \text{ kg}$, estimated age $t_{\text{ring}} = 45.0 \text{ Myr}$, and ring rain inflow $\dot{M}_{\text{rain}} = 4800 \text{ kg/s}$ (Iess et al. 2019, Zhang et al. 2017) with $R^2 = 0.9998$.